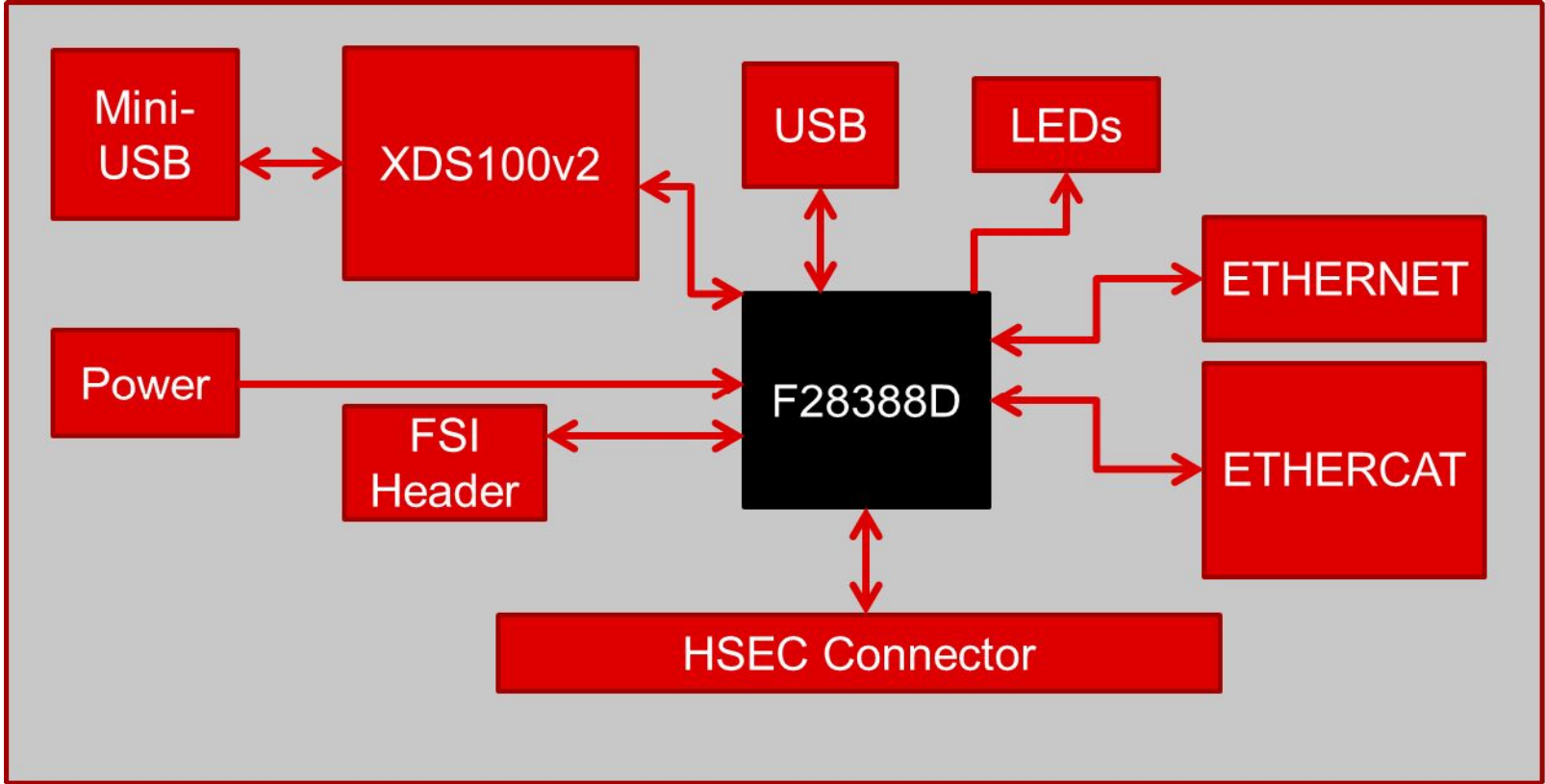
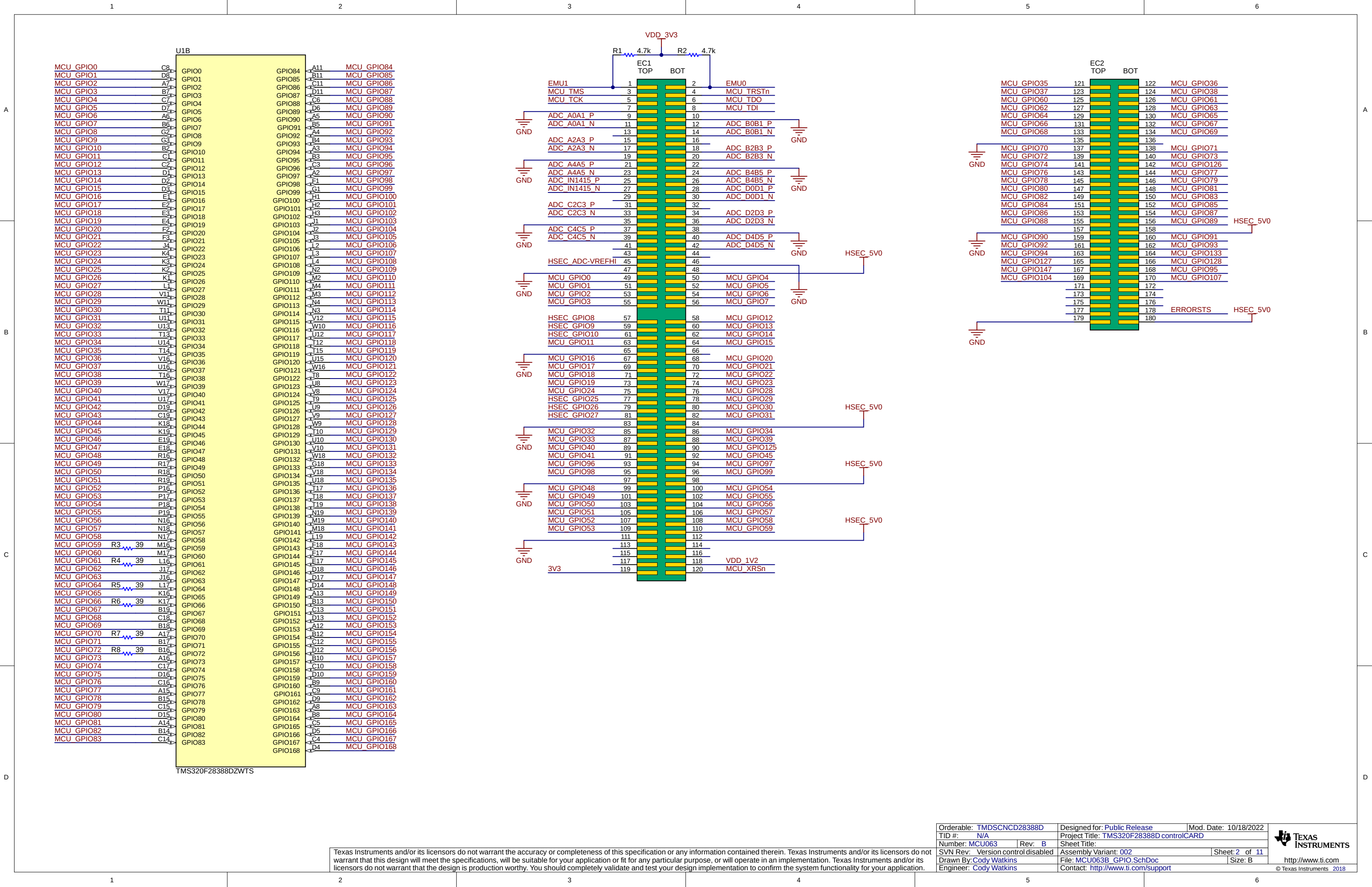


- 1) USB Differential Pairs - 90 Ohm
(A)USB_N and USB_P
(B)MCU_GPIO42 and MCU_GPIO43
- 2)ADC Differential pair Impedance Matching
(A)HSEC_ADC even pins should match with HSEC_ADC + 1 pin(ie HSEC_ADC-C2 should match with HSEC_ADC-C3)
(B)MCU_ADC even pins should match with MCU_ADC + 1 pin(ie MCU_ADC-A0 should match with MCU_ADC-A1)
- 3)ETHERNET and ETHERCAT Differential pairs
(A)TD_P and TD_N
(B)RD_P and RD_N

Revision History				
Rev	ECN #	Approved Date	Approved by	Notes
E1	N/A	N/A	N/A	Initial
E2	N/A	N/A	N/A	Fixed Ethernet MDC Bug in PCB Replaced U20 with an inverting buffer
A	N/A	N/A	N/A	Public Release
B	N/A	N/A	N/A	Changed Default clock to 25MHz Added 1V2, 3V3, and ERRORSTS to HSEC connector Enhanced J2 to support 2 data lines

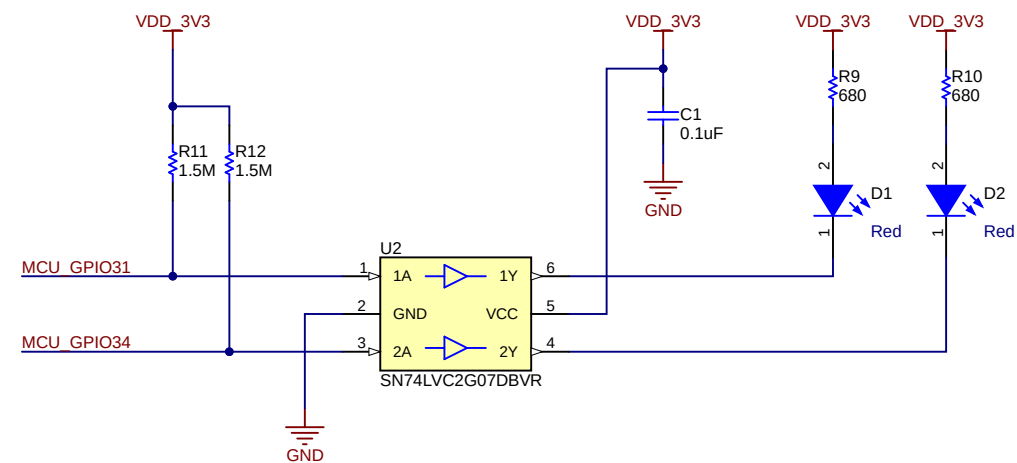
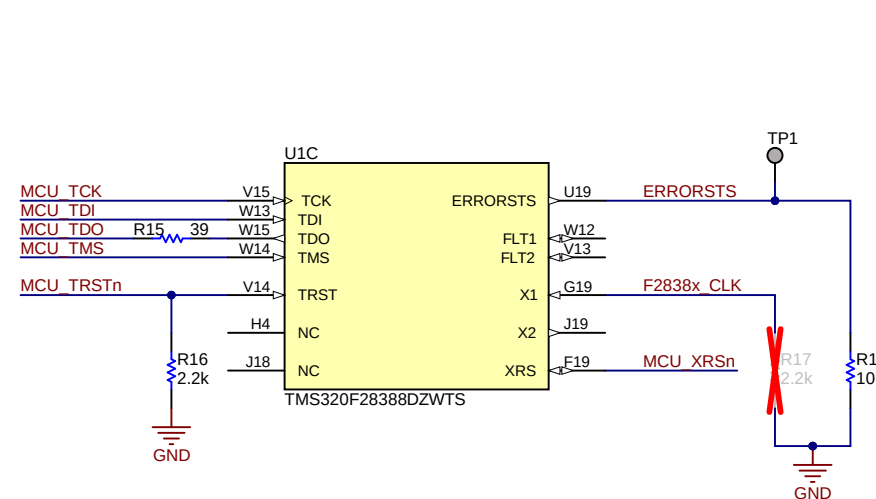


This controlCARD does not support the High Density(HD) connector, or EMIF, as previously supported by TMDSCNCDF28379D.

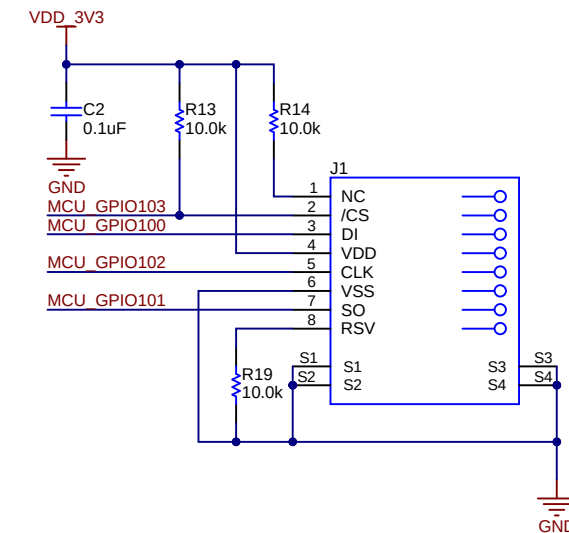


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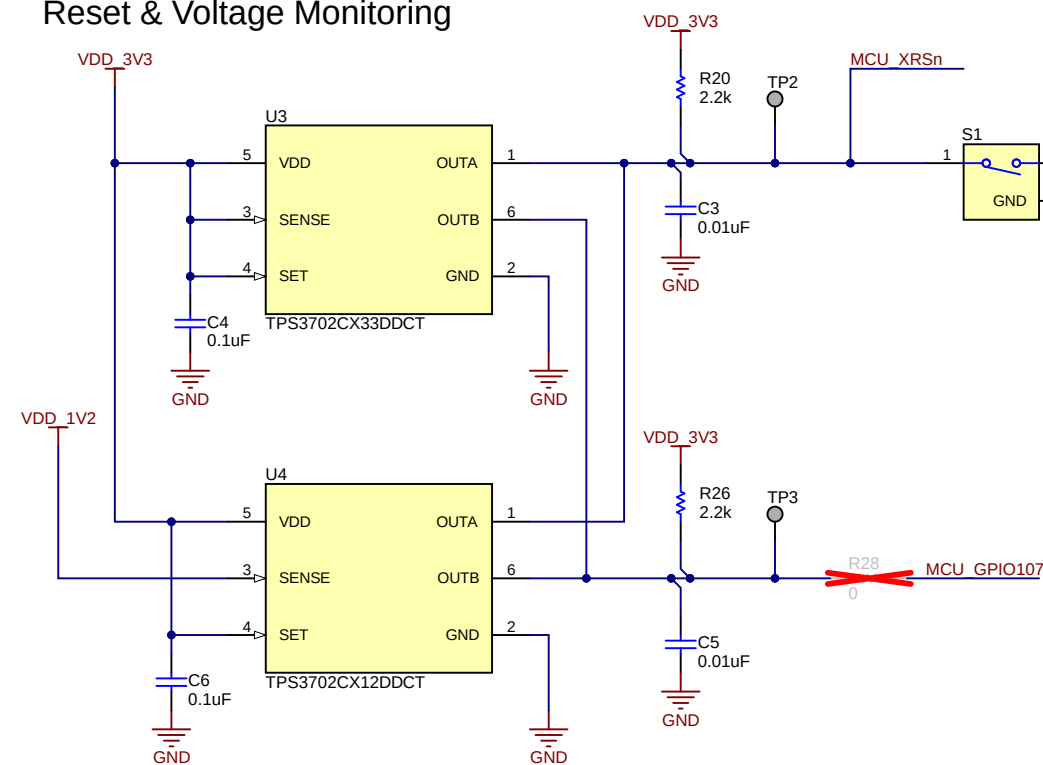
Orderable: TMDSCNCD28388D	Designed for: Public Release	Mod. Date: 10/18/2022
TID #: N/A	Project Title: TMS320F28388D controlCARD	
Number: MCU063	Rev: B	Sheet Title:
SVN Rev: Version control disabled	Assembly Variant: 002	Sheet: 2 of 11
Drawn By: Cody Watkins	File: MCU063B_GPIO.SchDoc	Size: B
Engineer: Cody Watkins	Contact: http://www.ti.com/support	



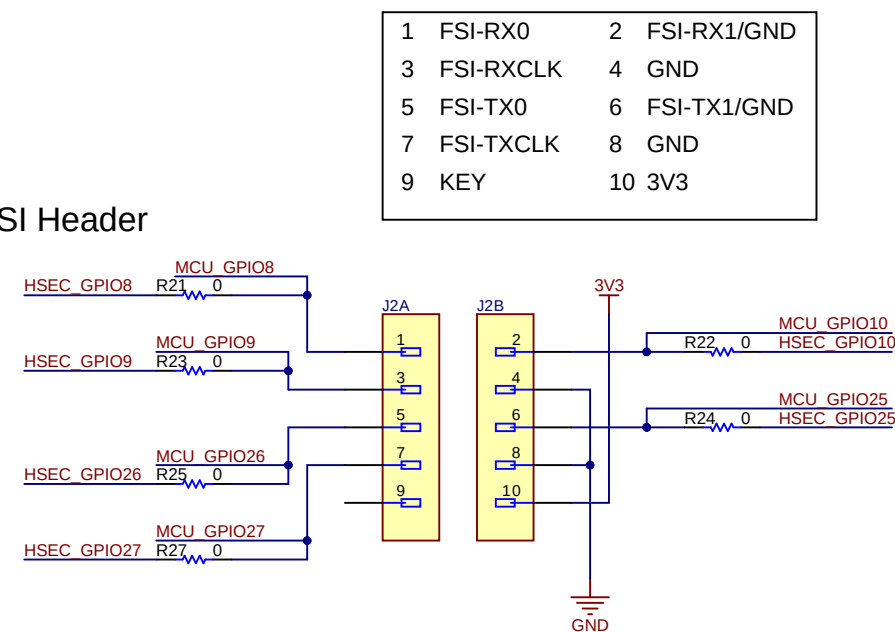
Micro-SD Card



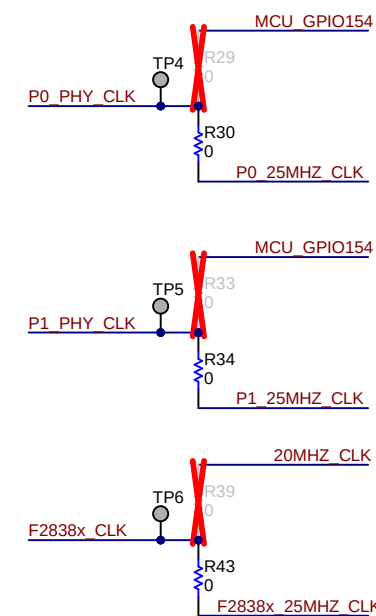
Reset & Voltage Monitoring



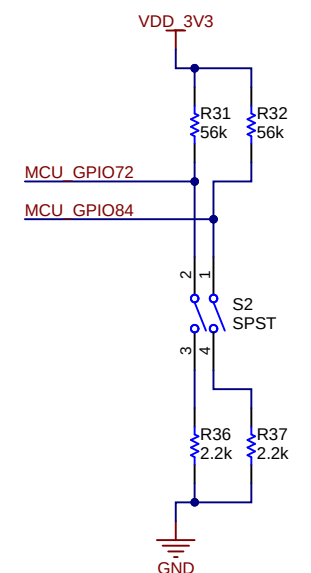
FSI Header



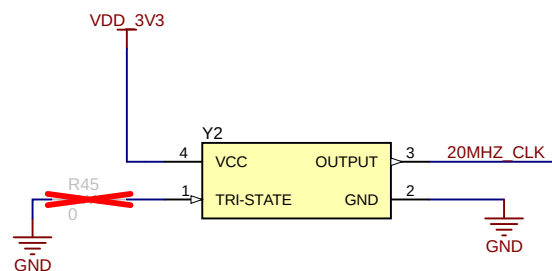
Clock Source Selection



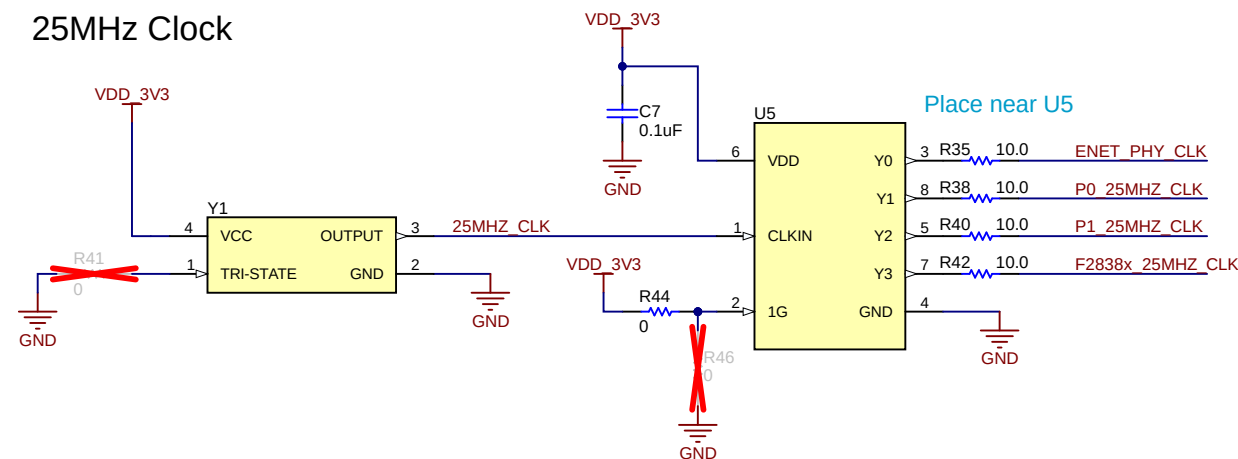
Boot Mode Selection Switch

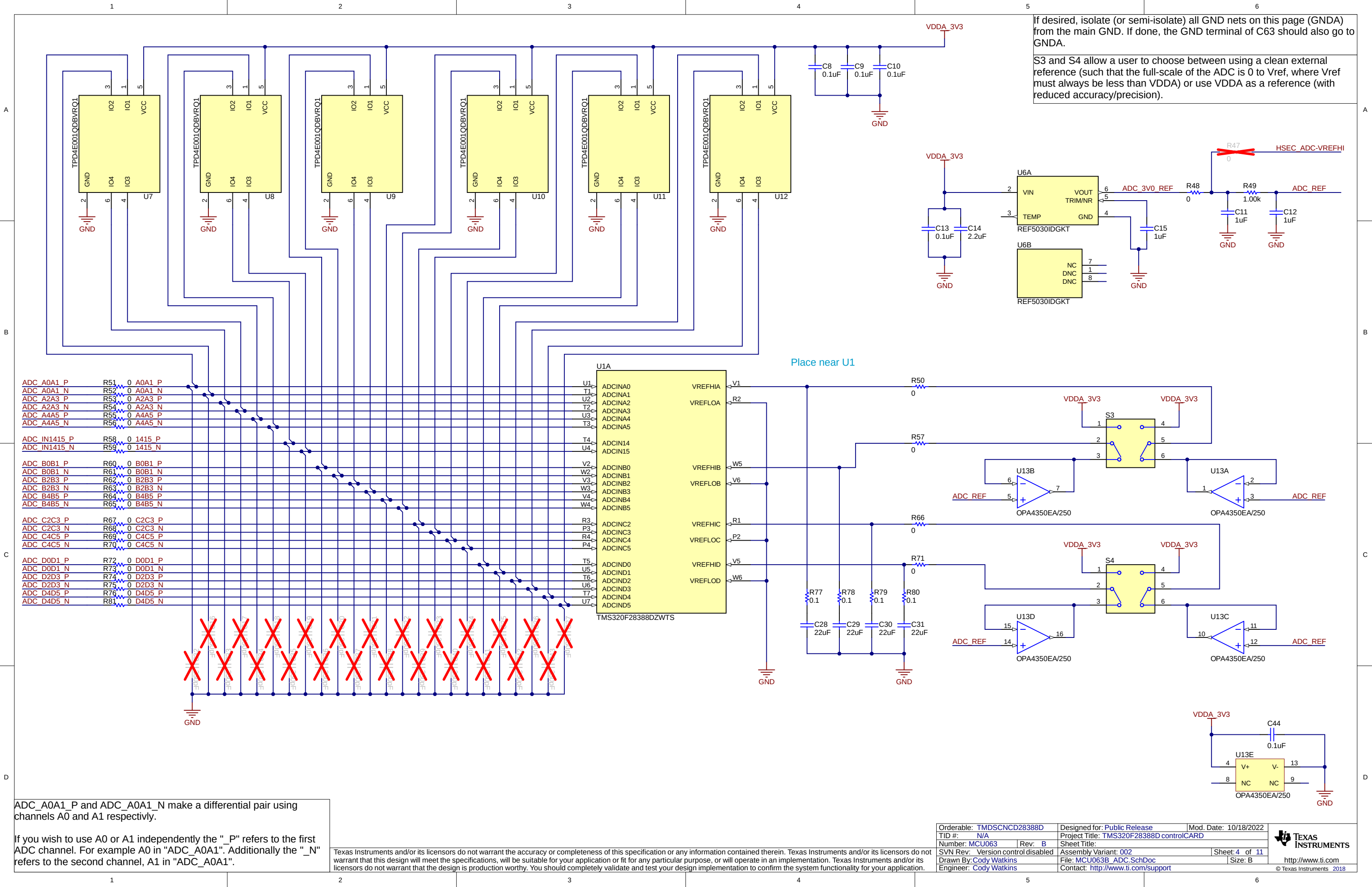


20MHz Clock



25MHz Clock





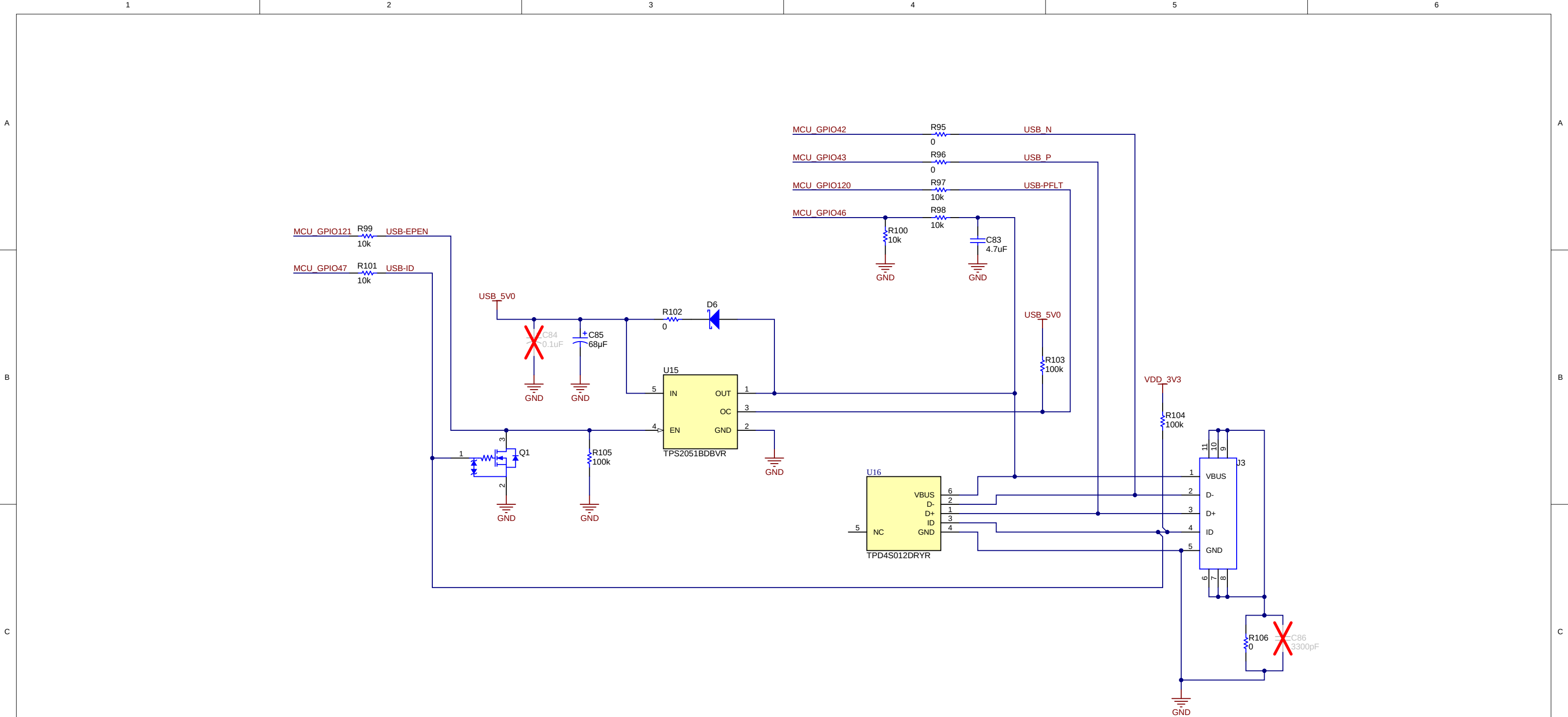
If desired, isolate (or semi-isolate) all GND nets on this page (GNDA) from the main GND. If done, the GND terminal of C63 should also go to GNDA.

S3 and S4 allow a user to choose between using a clean external reference (such that the full-scale of the ADC is 0 to Vref, where Vref must always be less than VDDA) or use VDDA as a reference (with reduced accuracy/precision).

ADC_A0A1_P and ADC_A0A1_N make a differential pair using channels A0 and A1 respectively.

If you wish to use A0 or A1 independently the "_P" refers to the first ADC channel. For example A0 in "ADC_A0A1". Additionally the "_N" refers to the second channel, A1 in "ADC_A0A1".

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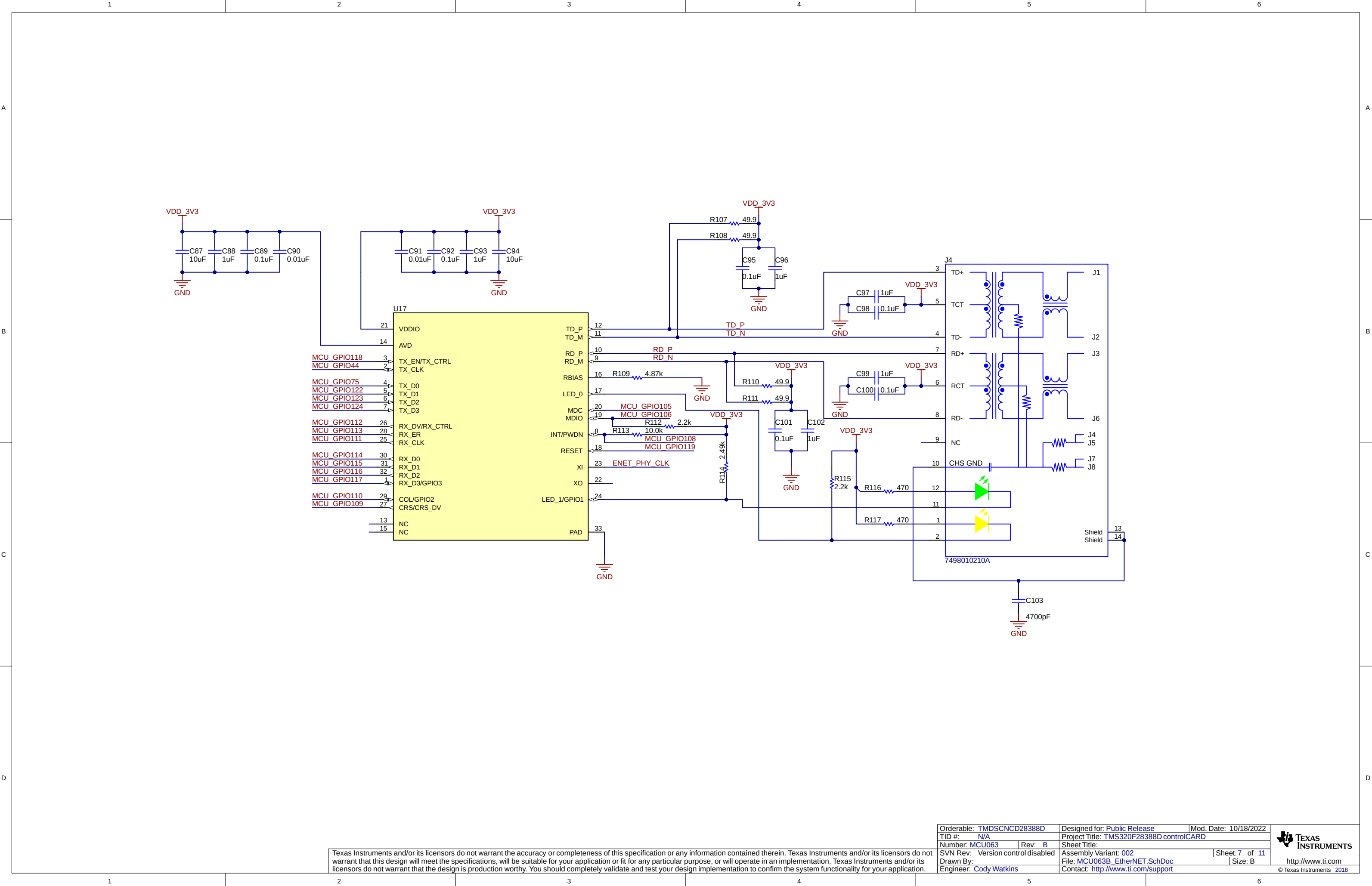


NOTE: USB VBUS, ID, PFLT & EPEN do not have a specific mux position in this device.

In this controlCARD, a standard GPIO is used to detect changes to these signals.

R102 is populated to allow a customer to evaluate the controlCARD without a baseboard if they desire. USB specifies the total bus capacitance on any device as 10uF max. R102 should be depopulated in most end applications.

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TID #: N/A	Project Title: TMS320F28388D controlCARD	
Number: MCU063	Rev: B	Sheet Title:
SVN Rev: Version control disabled	Assembly Variant: 002	Sheet: 6 of 11
Drawn By: Cody Watkins	File: MCU063B_USB.SchDoc	Size: B
Engineer: Cody Watkins	Contact: http://www.ti.com/support	



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TID #: N/A	Project Title: TMS320F28388D controlCARD	
Number: MCU063	Rev: B	Sheet Title:
SVN Rev: Version control disabled	Assembly Variant: 002	Sheet: 7 of 11
Drawn By:	File: MCU063B EtherNET.SchDoc	Size: B
Engineer: Cody Watkins	Contact: http://www.ti.com/support	

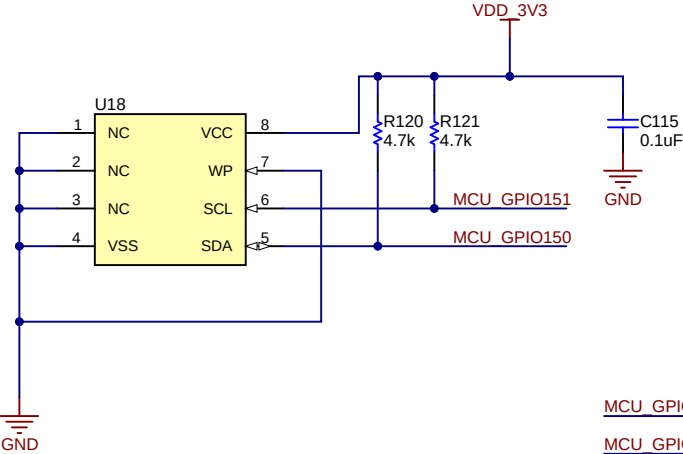


TEXAS
INSTRUMENTS

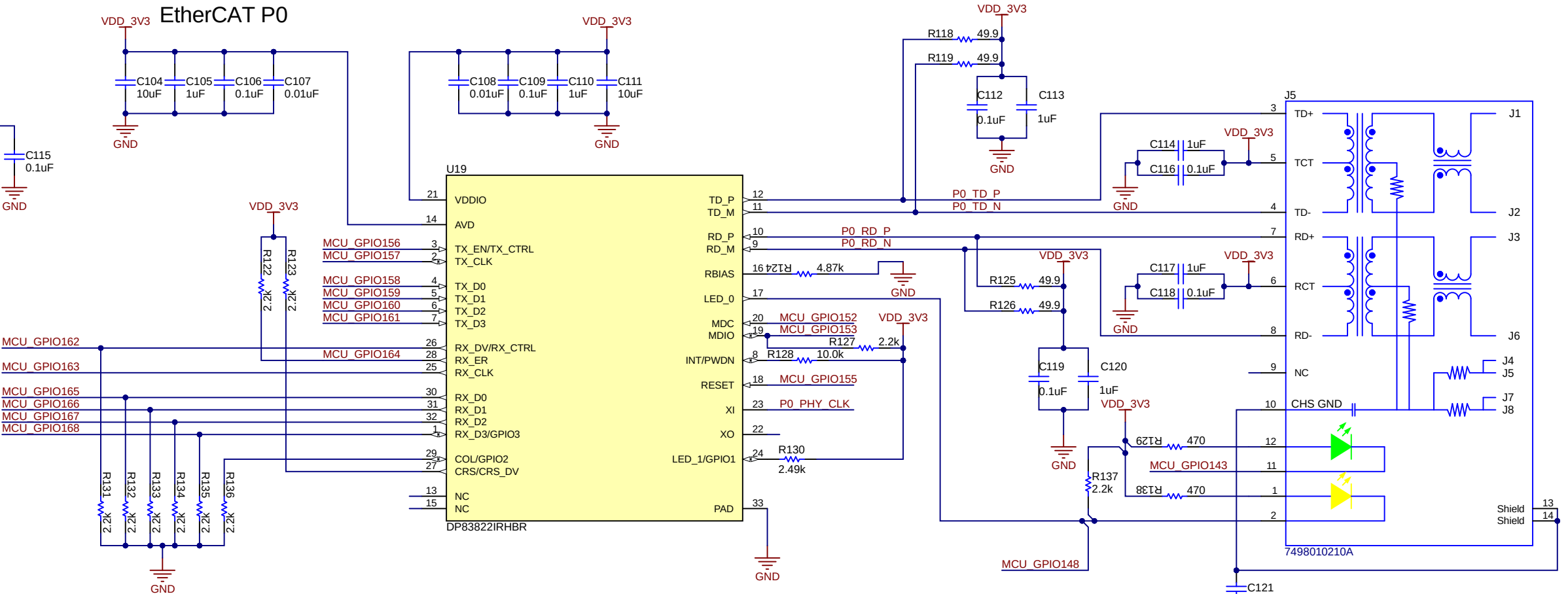
<http://www.ti.com>

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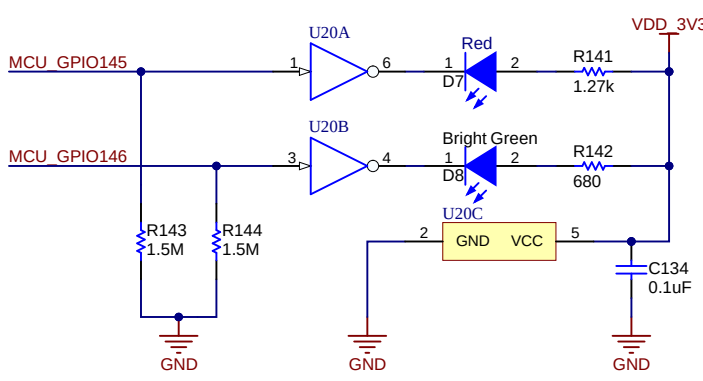
EtherCAT EEPROM



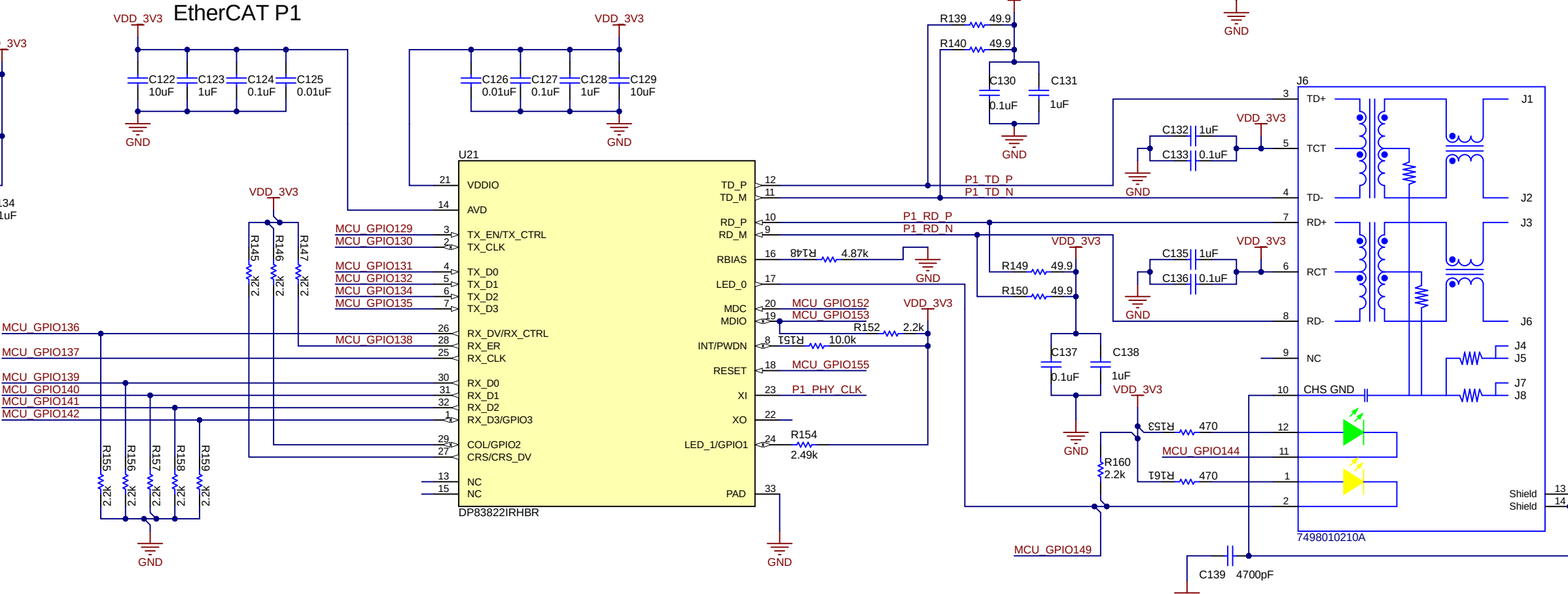
EtherCAT P0



Run and Error LEDs

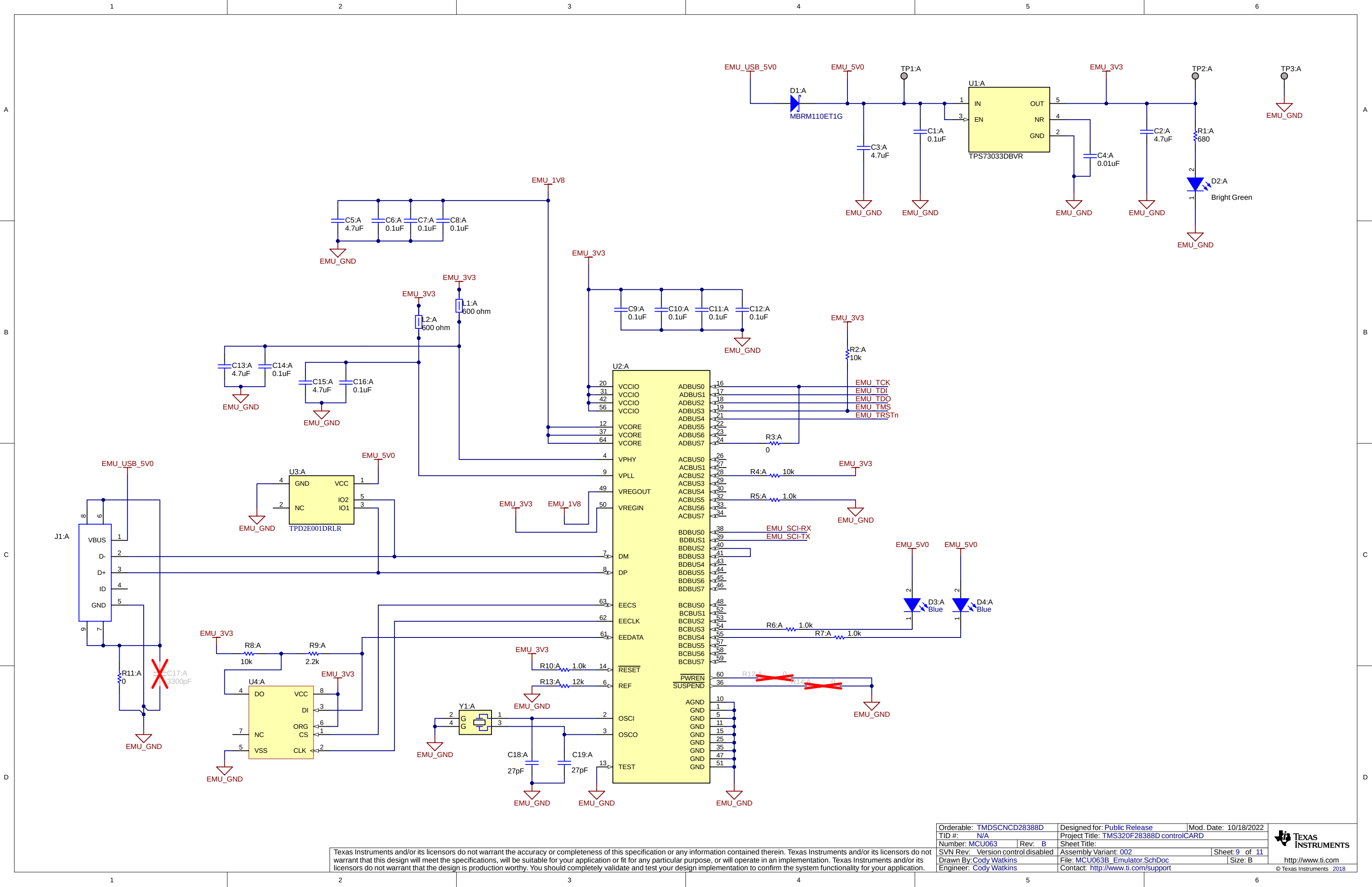


EtherCAT P1



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Orderable: TMDSCNCD28388D	Designed for: Public Release	Mod. Date: 10/18/2022
TID #: N/A	Project Title: TMS320F28388D controlCARD	
Number: MCU063	Rev: B	Sheet Title:
SVN Rev: Version control disabled	Assembly Variant: 002	Sheet: 8 of 11
Drawn By:	File: MCU063B EtherCAT.SchDoc	Size: B
Engineer: Cody Watkins	Contact: http://www.ti.com/support	



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TID #: N/A	Project Title: TMS320F28388D controlCARD	
Number: MCU063	Rev: B	Sheet Title:
SVN Rev: Version control disabled	Assembly Variant: 002	Sheet: 9 of 11
Drawn By: Cody Watkins	File: MCU063B_Emulator.SchDoc	Size: B
Engineer: Cody Watkins	Contact: http://www.ti.com/support	

A

B

C

D

A

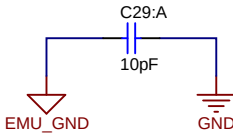
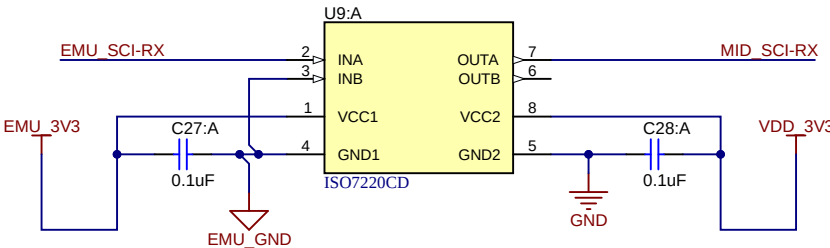
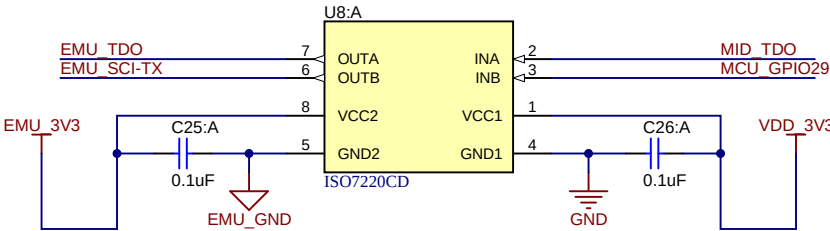
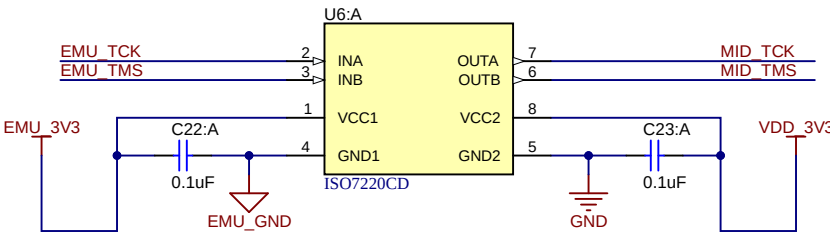
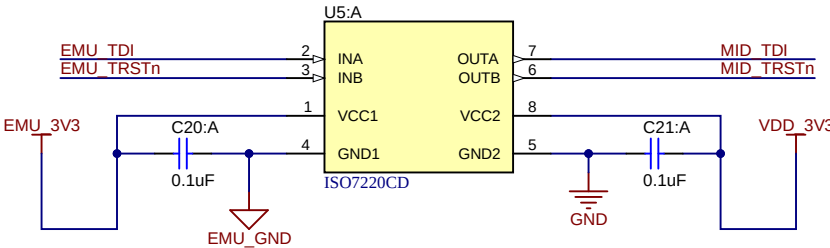
B

C

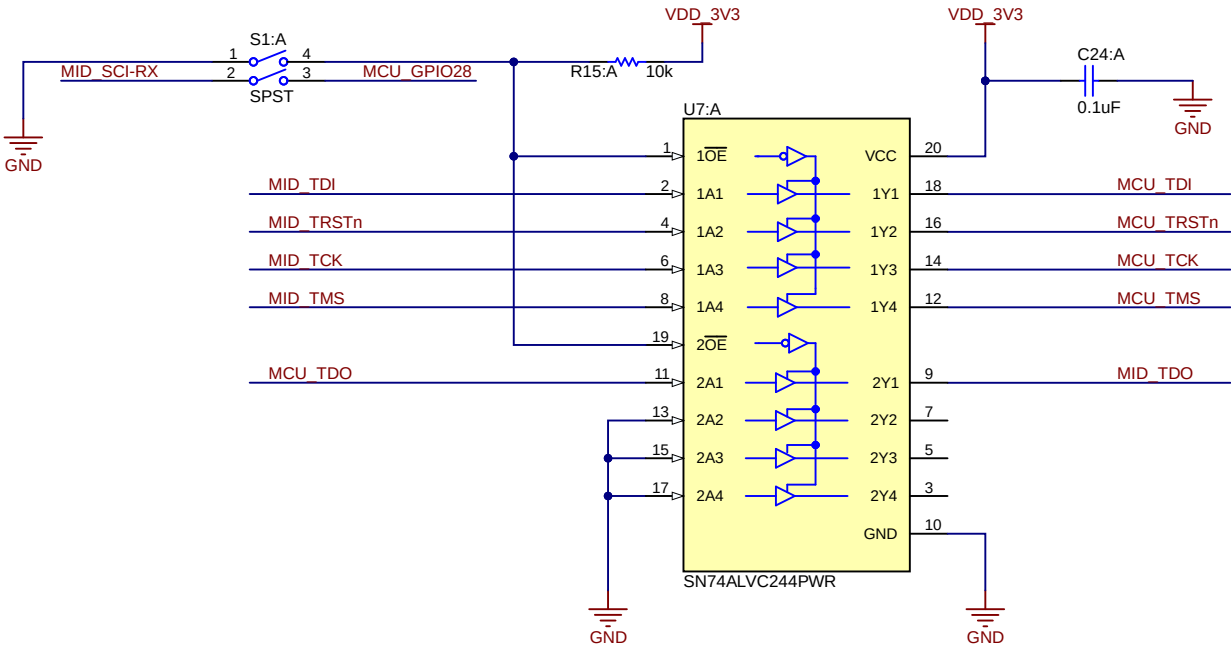
D

S1:A - Emulation & GPIO28 Switch

- POS 1 ON: Use xds100v2 emulator that is on the cCARD
- POS 1 OFF: Boot from FLASH/peripheral (see boot mode switch) OR use emulator on baseboard
- POS 2 ON: GPIO28 will be controlled by the USB-to-UART adapter on the FTDI chip
- POS 2 OFF: GPIO28 can be controlled by a pin in HSEC connector



WARNING: To avoid potential shock hazard in a high-voltage setting, it is strongly recommended to remove the Y cap (C29:A) from the EVM.



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TID #: N/A	Project Title: TMS320F28388D controlCARD	
Number: MCU063	Rev: B	Sheet Title:
SVN Rev: Version control disabled	Assembly Variant: 002	Sheet: 10 of 11
Drawn By: Cody Watkins	File: MCU063B Emulator interface.SchDoc	Size: B
Engineer: Cody Watkins	Contact: http://www.ti.com/support	



PCB Number: MCU063
PCB Rev: B

PCB
LOGO
Texas Instruments



PCB
LOGO
FCC disclaimer

PCB
LOGO
WEEE logo

PCB
LOGO
ETHERCAT LABEL

LBL1

PCB Label

THT-14-423-10
Size: 0.65" x 0.20 "

ZZ1

Label Assembly Note

This Assembly Note is for PCB labels only

ZZ2

Assembly Note

These assemblies are ESD sensitive, ESD precautions shall be observed.

ZZ3

Assembly Note

These assemblies must be clean and free from flux and all contaminants. Use of no clean flux is not acceptable.

ZZ4

Assembly Note

These assemblies must comply with workmanship standards IPC-A-610 Class 2, unless otherwise specified.

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Orderable: TMDSCNC28388D	Designed for: Public Release	Mod. Date: 10/17/2022
TID #: N/A	Project Title: TMS320F28388D controlCARD	
Number: MCU063	Rev: B	Sheet Title:
SVN Rev: Version control disabled	Assembly Variant: 002	Sheet: 11 of 11
Drawn By:	File: MCU063B EVM Hardware.SchDoc	Size: B
Engineer: Cody Watkins	Contact: http://www.ti.com/support	